DO YOU KNOW?

Meningitis is an inflammation of the meninges. It can occur when fluid surrounding the meninges becomes infected. The most common causes of meningitis are viral and bacterial infections.

A human brain is formed of three main parts: (a) forebrain (b) midbrain and (c) hindbrain.

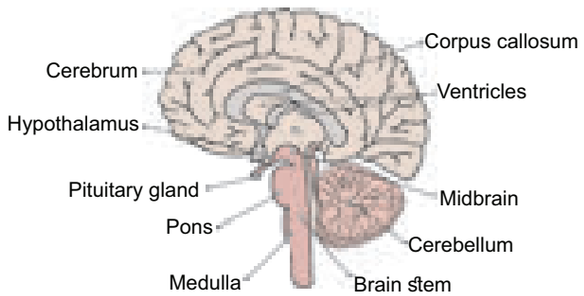


Fig. 15.5 L.S of Human Brain

Forebrain: The forebrain is formed of **cerebrum** and **diencephalon**. The latter consists of dorsal *thalamus* and ventral *hypothalamus*.

Cerebrum

It is the largest portion forming nearly two-third of the brain. The cerebrum is longitudinally divided into two halves as right and left **cerebral hemispheres** by a deep cleft called **median cleft**. Two cerebral hemispheres are interconnected by thick band of nerve fibres called **corpus callosum.** The outer portion of each cerebral hemisphere is formed of grey matter and is called **cerebral cortex.** The inner or deeper part is formed of white matter and is called **cerebral medulla.** The cortex is extremely folded forming elevations called **gyri** with depressions between them termed as **sulci** that increase its surface area.

Each cerebral hemisphere is divisble into a frontal lobe, a parietal lobe, a temporal lobe and an occipital lobe. These lobes are also known as **cerebral lobes** and are associated with specific functions. Any damage in specific lobe inturn affects its function.

The cerebrum is responsible for the thinking, intelligence, consciousness, memory, imagination, reasoning and willpower.

Thalamus

Thalamus present in cerebral medulla is a major conducting centre for sensory and motor signalling. It acts as a **relay centre.**

Hypothalamus

It lies at the base of the thalamus. It controls involuntary functions like hunger, thirst, sleep, sweating, sexual desire, anger, fear, water balance, blood pressure etc. It acts as a **thermoregulatory** (temperature control) **center** of the body. It controls the secretion of hormones from anterior pituitary gland and is an important link between nervous system and endocrine system.

Midbrain

It is located between thalamus and hind brain. The dorsal portion of the mid brain consists of four rounded bodies called **corpora quadrigemina** that control visual and auditory (hearing) reflexes.

Hindbrain

It is formed of three parts **cerebellum, pons** and **medulla oblongata**.

DO YOU KNOW?

The human brain constitutes nearly 60 percent of fat. The most crucial molecules that determine our brain's integrity and the ability are Essential Fatty Acids (EFAs). EFAs cannot be synthesised and must be obtained from food. Fish, green leafy vegetables, almond, walnut are rich sources of EFAs.

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